



AFT-Net: Adaptive Representation Learning for Robust Crowd Counting and Localization

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Abstract. Crowd counting and localization, fundamental computer vision tasks, hold significant promise for applications in public safety and intelligent retail. However, the performance of existing methods degrades drastically when faced with challenging real-world scenarios, particularly under complex indoor lighting conditions. This is primarily attributed to their reliance on fixed, hand-crafted representations, which cannot adapt to the dramatic appearance changes caused by glare and shadows. To address this issue, we propose a novel model named AFT-Net. At its core is an Adaptive Focal Inverse Distance Transform (AFT) mechanism, which dynamically learns the optimal representation function for individual locations via a Hyper-Network. Furthermore, we incorporate the Convolutional Block Attention Module (CBAM) to enhance the network's perception of critical regions. A key feature of our design is that the AFT module is only active during training to guide the learning process; it is discarded during inference, ensuring no extra computational cost. To systematically evaluate this problem, we construct and release a new Indoor-Robustness Benchmark dataset. Extensive experiments demonstrate that AFT-Net achieves state-of-the-art (SOTA) performance and superior robustness, fully validating its effectiveness.

Keywords: Crowd Localization · Robustness · Adaptive Representation · Attention Mechanism · Challenging Illumination

1 Introduction

In modern intelligent surveillance systems, accurately localizing every individual in a scene is a critical capability. Imagine a security system monitoring a busy subway station: a sudden bright glare from a reflection or a deep shadow cast by a pillar could momentarily blind a conventional computer vision model, causing it to miss individuals or generate false alarms. This is not just a theoretical problem; it is a significant barrier to the reliable deployment of crowd analysis technologies. Recent methods, such as FIDTM [1], have made great strides in localization accuracy under ideal conditions. However, their performance remains fragile in the face of real-world complexities like challenging illumination. We argue that the core reason for this fragility lies in a fundamental

design flaw: **Representation Rigidity**. Existing methods learn to regress a target map (e.g., a density map or a FIDT map) that is generated using a fixed, static formula. This *one-size-fits-all* approach inherently lacks the flexibility to handle the diverse and dynamic appearance changes caused by factors like glare and shadows. The ideal representation for a brightly lit, sparse scene should fundamentally differ from that of a dark, dense scene. To overcome this limitation, we propose AFT-Net, a model that embraces adaptivity. Our key contributions are:

- We introduce an **Adaptive Focal Transform (AFT)** mechanism. Instead of using a fixed target representation, our model employs a hyper-network to dynamically generate a scene-specific learning target during training.
- We integrate a **CBAM attention module** to work in synergy with AFT, refining features to better inform the adaptive process.
- We construct and will release the first **Indoor-Robustness Benchmark**, a dataset specifically designed to evaluate performance under challenging lighting conditions, comprising `indoor_base`, `indoor_light`, and `indoor_shadow` subsets.
- Through extensive experiments, we demonstrate that AFT-Net not only achieves highly competitive performance but also exhibits significantly enhanced robustness against adverse lighting.

2 Related Work

2.1 Localization-Based Crowd Counting and Localization

To overcome the localization ambiguity of traditional density map methods, researchers have explored methods that directly locate each individual. FIDTM [1] is a milestone work that significantly improves localization accuracy by predicting a novel Focal Inverse Distance Transform map. P2PNet [4] identifies individuals by predicting each point and its nearest neighbor. However, the target representations these methods rely on are generated based on fixed, man-made rules. This characteristic forms a natural bottleneck to their adaptability, which is the main motivation for our work.

2.2 Attention Mechanisms in Computer Vision

Attention mechanisms aim to mimic the human visual system, enabling neural networks to focus on more important parts of the input. From the pioneering SENet [5] to CBAM [3], attention has been proven as a universal and effective module for improving vision models. In our work, we leverage it to suppress noise caused by illumination and enhance the perception of valid information.

2.3 Common Crowd Counting Datasets

The field has been advanced by several key benchmarks. ShanghaiTech [7] provides dense (Part A) and relatively sparse (Part B) scenes. UCF-QNRF [12] and

JHU-CROWD++ [6] pushed the scale with higher crowd counts and greater diversity. While these datasets are invaluable for general-purpose evaluation, they do not specifically isolate or systematically test for robustness against challenging lighting conditions. This gap motivates our development of the Indoor-Robustness Benchmark, which is tailored to evaluate model performance specifically under varied and adverse illumination.

3 Proposed Method

3.1 Overall Architecture

The overall architecture of our proposed AFT-Net is illustrated in Fig. 1. The model is built upon a shared backbone, HRNet-W48 [2], which is first enhanced with a **Convolutional Block Attention Module (CBAM)** [3]. The CBAM module refines the features extracted by the backbone by sequentially applying channel and spatial attention, allowing the network to focus on more informative features and suppress irrelevant ones, which is particularly crucial under complex lighting. The refined features are then processed by two parallel branches:

- **Prediction Head:** This is the main branch, active during both training and inference. It is responsible for decoding the features and generating the final localization map (M_{pred}).
- **AFT Head:** This is our core innovation, a guidance module that is **only active during the training phase**. It contains a hyper-network that analyzes the scene’s global context and predicts adaptive parameters (α, β) .

This dual-path design allows the AFT Head to act as a *teacher* module, guiding the backbone to learn robust, adaptive representations without incurring any computational overhead during inference, as it is completely discarded.

3.2 Adaptive Focal Transform (AFT)

This section details the working process of our AFT mechanism. The core idea is to transform the learning paradigm from fitting a fixed target to a dynamic one, where the target representation is tailored for each specific input image.

Motivation. Traditional FIDTM uses a fixed function, $F_{fix}(p) = 1/(d(p) + \alpha)$, where α is a global hyperparameter. While effective in sparse scenes, this rigidity leads to significant problems in dense crowds. As illustrated in Fig. 2, the standard FIDT map (b) generates blurry, overlapping responses where individuals are tightly packed. These smeared peaks make it extremely difficult for a simple local maxima search to accurately distinguish and localize each person, leading to poor localization performance. This motivates the need for a representation that can adapt its structure based on scene density, which is the core goal of our AFT mechanism. The result from our method (c) shows distinct, sharp peaks even in dense regions, demonstrating the effectiveness of our adaptive approach.

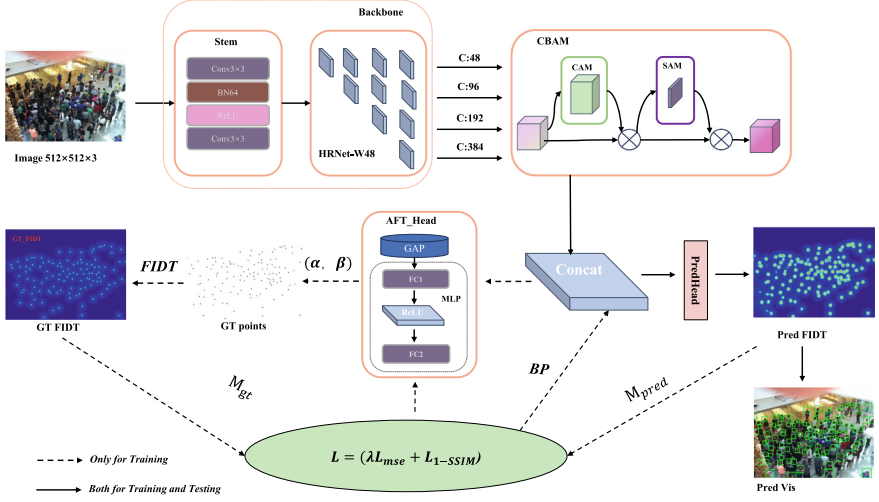


Fig. 1. The detailed architecture of AFT-Net, illustrating the dual-path design. The shared backbone feeds features into two parallel heads. The Prediction Head (Path A) is active for both training and inference. The AFT Head (Path B) is a guidance module used only during training to generate adaptive parameters, and is discarded for inference, ensuring high efficiency.

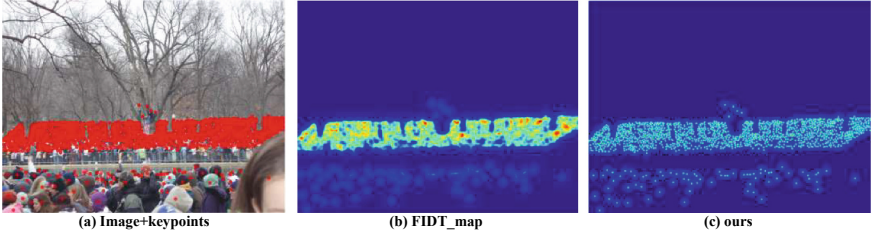


Fig. 2. Visual motivation for our Adaptive Focal Transform (AFT). (a) Input image with ground-truth keypoints in a dense crowd. (b) The map generated by the baseline FIDTM, where responses for individuals merge into a single, blurry region. (c) Our AFT-Net produces sharp, distinguishable peaks, enabling accurate localization even in dense scenarios.

AFT Mechanism. We propose a more flexible transformation form:

$$F_{adp}(p) = \frac{1}{d(p)^\beta + \alpha} \quad (1)$$

where the parameters (α, β) are dynamically generated by a light-weight **Hyper-Network** H (implemented as a small MLP), which constitutes the AFT Head. We designed this formulation because β and α play distinct and critical roles. The parameter β , as the exponent of the distance term $d(p)$, effectively controls the **sharpness** of the response peaks. In dense crowds where distances $d(p)$ are

small, a larger β can amplify minor distance differences, making it possible to produce distinguishable, sharp peaks for nearby individuals and thus resolving the blurry response issue shown in Fig. 2(b). Meanwhile, the parameter α primarily acts as a **global offset**, adjusting the baseline of the response map. This is crucial for suppressing background noise and adapting to the overall brightness of different scenes. The dynamic combination of these two parameters provides the model with sufficient flexibility to generate an optimal target representation. The process is as follows:

1. **Scene Feature Encoding:** The shared backbone (HRNet + CBAM) extracts a feature map F_{att} .
2. **Global Context Analysis:** The AFT Head applies Global Average Pooling (GAP) to F_{att} to obtain a compact global context vector v .
3. **Parameter Prediction:** The vector v is then fed into the Hyper-Network H to regress the optimal parameter pair (α, β) .

$$(\alpha, \beta) = H(\text{GAP}(F_{att})) \quad (2)$$

End-to-End Training via Implicit Supervision. The Hyper-Network is trained implicitly through a unified, combined loss function. For each training step, the predicted parameters (α, β) are used with the ground-truth point annotations to dynamically construct an **adaptive ground-truth map**, M_{gt} . The final loss is then calculated between the prediction M_{pred} and this adaptive target M_{gt} :

$$L = L_{1-SSIM} + \lambda \cdot L_{MSE} \quad (3)$$

where $L_{1-SSIM} = 1 - \text{SSIM}(M_{pred}, M_{gt})$ focuses on preserving the local peak structure for accurate localization, and L_{MSE} ensures numerical consistency for accurate counting. λ is a balancing hyperparameter. The gradient from this loss implicitly supervises the Hyper-Network, guiding it to generate parameters that define an optimal learning target and thus lead to a lower overall loss.

4 Experiments

4.1 Experimental Setup

Implementation Details. Our experimental setup is built upon the HRNet-W48 backbone, with weights pre-trained on ImageNet. Training and evaluation are performed on a workstation equipped with two NVIDIA RTX 4090 GPUs, each providing 24 GB of video memory. For optimization, we employ the Adam optimizer. The learning rate is initialized to 1×10^{-4} and is subject to a step decay schedule, where it is reduced by a factor of 0.5 every 500 epochs.

Hyper-Network Architecture. Our Hyper-Network H is a lightweight Multi-Layer Perceptron (MLP) consisting of two fully-connected (FC) layers. The input is the global context vector obtained from Global Average Pooling (GAP). The first FC layer maps this vector to a 128-dimensional hidden layer, followed by a ReLU activation function. The second FC layer outputs two scalar values for the parameters (α, β) . To ensure training stability, we use a scaled Sigmoid function to constrain the output ranges, limiting α to $(0, 0.1)$ and β to $(0.5, 1.0)$.

The Indoor-Robustness Benchmark. To rigorously evaluate model robustness against real-world lighting variations, we constructed and will release a new benchmark, the **Indoor-Robustness Benchmark**. Unlike benchmarks that use generic indoor scenes, our goal was to create a dataset focused specifically on these challenges. **indoor_base**: This is our foundational dataset, for which we meticulously curated images from the JHU-CROWD++ [6] and ShanghaiTech [7] datasets. We specifically selected scenes that already exhibit complex natural illumination, such as strong directional light, reflections, and diverse shadow patterns. This process resulted in a dataset comprising 1334 images for training, 285 for validation, and 325 for testing. Models trained on this set are expected to learn features that are inherently more resilient to lighting variations. **indoor_light and indoor_shadow**: To specifically test model performance under extreme conditions not fully covered by the base set, we generated two synthetic test sets by applying augmentation algorithms to the 325 test images from **indoor_base**.

- The **indoor_light** set is created by simulating realistic lens flare and glare. We model the glare effect $G(p)$ at each pixel p by generating several polygonal *ghosts* (randomly between 3 and 7) and a radial glow originating from a randomized light source position p_s placed just outside the image frame. The final pixel intensity $I_{out}(p)$ is a blend of the original pixel $I_{in}(p)$ and the glare effect:

$$I_{out}(p) = (1 - \lambda) \cdot I_{in}(p) + \lambda \cdot G(p) \quad (4)$$

where the glare intensity λ is randomly sampled from the range $[0.2, 0.4]$.

- The **indoor_shadow** set is generated by overlaying synthetic soft-edged shadows. We first create a shadow mask $M_s(p)$ by generating a random polygonal shape and then applying a strong Gaussian blur with a kernel size of $(101, 101)$ to soften the edges. The final image is produced by a multiplicative darkening effect:

$$I_{out}(p) = I_{in}(p) \times (1 - \gamma \cdot M_s(p)) \quad (5)$$

where $M_s(p) \in [0, 1]$ is the value of the blurred mask at pixel p and the shadow's opacity γ is randomly sampled from $[0.4, 0.7]$.

Evaluation Metrics. We evaluate performance from two perspectives: counting and localization. For **counting evaluation**, we use the standard Mean Absolute Error (MAE) and Mean Squared Error (MSE). For a set of N test images, where

C_i is the ground-truth count and $\hat{C}_i$ is the predicted count for the i -th image, the metrics are defined as:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |C_i - \hat{C}_i| \quad (6)$$

$$\text{MSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (C_i - \hat{C}_i)^2} \quad (7)$$

For **localization evaluation**, we use Precision (P), Recall (R), and F1-score (F1). These metrics depend on matching predicted points to ground-truth points. A predicted point $\hat{p}_j$ is considered a True Positive (TP) if it is matched with a ground-truth point p_k that has not already been matched. The matching is determined by a distance threshold σ . Specifically, a match occurs if the Euclidean distance between the points is within σ :

$$\text{Distance}(\hat{p}_j, p_k) = \sqrt{(\hat{x}_j - x_k)^2 + (\hat{y}_j - y_k)^2} \leq \sigma \quad (8)$$

Following standard practice in crowd analysis, We chose $\sigma = 4$ and $\sigma = 8$. We report results for multiple thresholds to provide a comprehensive assessment.

4.2 Ablation Study

To verify the effectiveness of each component, we conducted ablation studies on our **indoor_base** dataset. The quantitative results are presented in Table 1, with a corresponding performance visualization in Fig. 3. Our analysis reveals a clear progression:

- *Baseline vs. CBAM*: Simply adding CBAM to the baseline (*Baseline+CBAM*) provides a solid improvement across the board. MAE decreases from 86.2 to 82.3, and the F1-score ($\sigma = 8$) rises from 62.5% to 65.8%. This confirms that enhancing feature perception is beneficial.
- *The Role of Adaptivity*: The most significant leap in performance comes from introducing our adaptive mechanism. The model *AFT-Net (adaptive, w/o CBAM)* substantially outperforms the baseline. It's crucial to note that simply using our proposed formula with fixed parameters (*AFT-Net (fixed)*) offers no real benefit over the baseline. This decisively proves that the performance gain is driven by the dynamic, scene-specific adaptation, not just the transform's mathematical form.
- *Synergistic Effect*: The full *AFT-Net* model, combining CBAM and the AFT head, achieves the best results by a significant margin. As Fig. 3 illustrates, it consistently represents the peak performance, reducing MAE to 79.3 and boosting the F1-score ($\sigma = 8$) to 72.2%. This demonstrates a powerful synergy where CBAM refines the input features, enabling the AFT head to predict more effective adaptive parameters, thus validating our overall design.

Table 1. Ablation study on the `indoor_base` test set. The results demonstrate the progressive improvements from our proposed components. *fixed* α, β denotes using our AFT formula but with fixed parameters ($\alpha = 0.02, \beta = 0.75$). Best results are in **bold**.

Model	Counting Error ↓		Localization Performance ↑					
	MAE	MSE	$\sigma = 4$			$\sigma = 8$		
			Precision(%)	Recall(%)	F1-score(%)	Precision(%)	Recall(%)	F1-score(%)
Baseline (FIDTM)	86.2	236.3	45.6	39.5	42.4	67.2	58.3	62.5
Baseline + CBAM	82.3	217.5	47.3	41.2	44.3	69.9	60.8	65.8
AFT-Net (fixed α, β , w/o CBAM)	85.5	231.5	46.7	41.0	43.8	68.3	59.0	64.2
AFT-Net (adaptive, w/o CBAM)	81.6	218.7	48.2	43.2	45.7	70.4	62.4	66.7
AFT-Net (full)	79.3	209.7	52.3	50.7	51.6	74.5	68.9	72.2

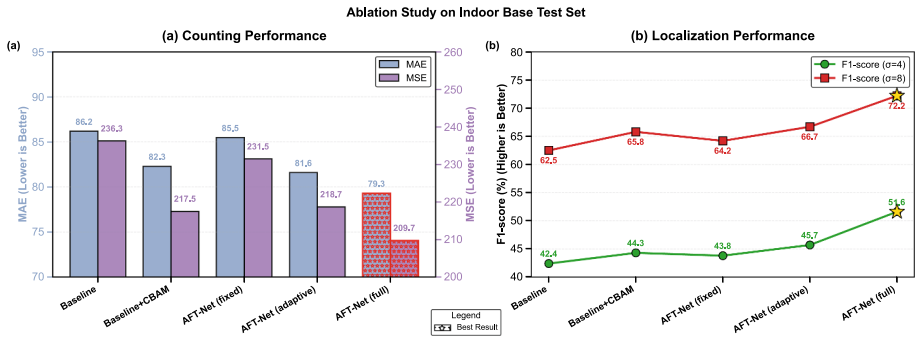


Fig. 3. Visual analysis of the ablation study on the `indoor_base` test set. (a) Counting performance, where lower MAE and MSE are better. (b) Localization performance, where a higher F1-score is better. The progressive improvements demonstrate the effectiveness of each added component, with the full AFT-Net model achieving the best results.

4.3 Illumination Robustness Analysis

We evaluated the models on all three test sets (base, light, and shadow) using weights trained solely on `indoor_base`. The results, presented in Table 2 and visualized in Fig. 4, show that AFT-Net possesses excellent robustness against varying illumination conditions. While the baseline model's performance degrades significantly under challenging lighting, our method maintains a strong level of performance.

Notably, although AFT-Net is superior in nearly all metrics, the baseline achieves a marginally better MSE on the `indoor_shadow` set (303.4 vs. 312.2). We hypothesize that this is because large, uniform dark patches in the synthetic shadows may cause rare, significant mispredictions, which are heavily penalized by the MSE metric. Nevertheless, our method's substantially better MAE and superior F1-score at $\sigma = 4$ on this set strongly confirm that its localization is more robust overall.

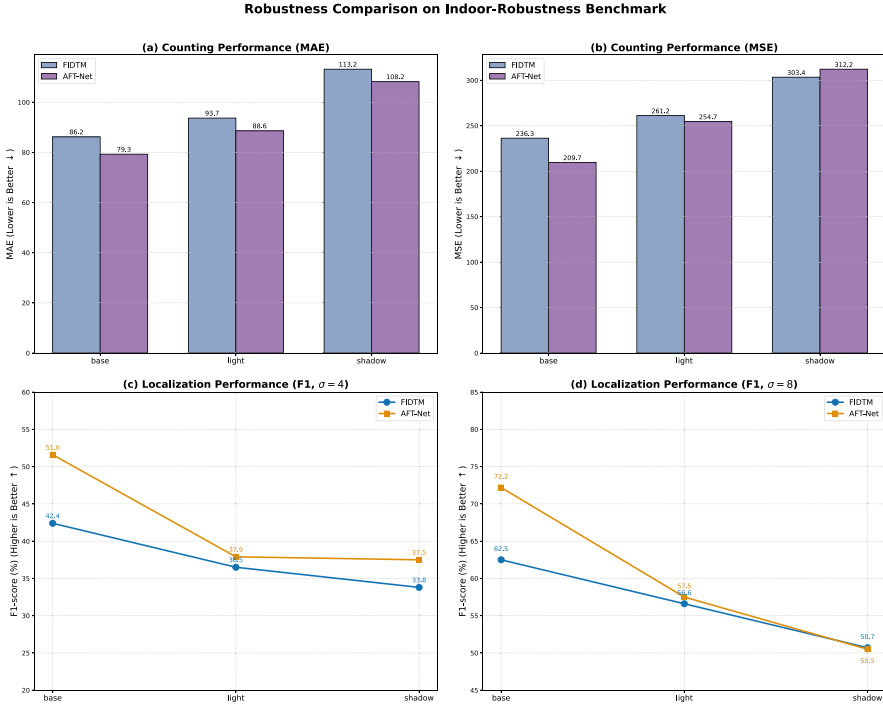


Fig. 4. Visual comparison of robustness between FIDTM and AFT-Net on our Indoor-Robustness Benchmark. (a) and (b) show counting performance (MAE and MSE), while (c) and (d) show localization performance (F1-score at $\sigma = 4$ and $\sigma = 8$). The results visually confirm the data in Table 2, highlighting AFT-Net’s superior stability under adverse lighting.

Table 2. Robustness comparison on the Indoor-Robustness Benchmark. Best results are in **bold**.

Test Set	Model	Counting Error ↓		Localization ($\sigma = 4$) ↑			Localization ($\sigma = 8$) ↑		
		MAE	MSE	Prec.(%)	Rec.(%)	F1(%)	Prec.(%)	Rec.(%)	F1(%)
base	FIDTM	86.2	236.3	45.6	39.5	42.4	67.2	58.3	62.5
	AFT-Net	79.3	209.7	52.3	50.7	51.6	74.5	68.9	72.2
light	FIDTM	93.7	261.2	39.2	33.5	36.5	60.7	50.6	56.6
	AFT-Net	88.6	254.7	40.8	35.7	37.9	59.4	53.2	57.5
shadow	FIDTM	113.2	303.4	37.4	30.1	33.8	56.5	44.8	50.7
	AFT-Net	108.2	312.2	40.5	34.6	37.5	55.9	45.8	50.5

4.4 Comparison with State-of-the-Art Methods

To comprehensively evaluate our method, we compare AFT-Net with representative state-of-the-art methods on four challenging benchmarks. The comparisons for counting performance are detailed in Table 3. The results show that AFT-Net

stands as a highly competitive method among the latest approaches. Specifically, on ShanghaiTech Part B, our method achieves the best MAE score (6.1), tying with P2R [23], and secures the second-best MSE (10.3). On the more challenging UCF-QNRF dataset, AFT-Net achieves the second-best results for both MAE (85.1) and MSE (141.2), demonstrating its strong generalization capabilities. While some recent methods like CLIP-EBC [22] show exceptional performance on JHU-CROWD++, our model still delivers strong, competitive results (66.5 MAE/250.8 MSE). This consistent high-ranking performance across multiple diverse datasets validates the effectiveness of our adaptive representation learning.

Table 3. Comparison of **counting performance (MAE/MSE)** with state-of-the-art methods on four public datasets. The best and second best results are in **bold** and underlined, respectively.

Method	Venue/Year	Backbone	ShanghaiTech A		ShanghaiTech B		UCF-QNRF		JHU-CROWD++	
			MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE
MCNN [7]	CVPR 2016	-	110.2	173.2	26.4	41.3	277.0	426.0	-	-
CSRNet [8]	CVPR 2018	VGG16	68.2	115.0	10.6	16.0	-	-	85.9	309.2
CAN [9]	CVPR 2019	VGG16	62.3	100.0	7.8	12.2	107.0	183.0	-	-
BL [10]	ICCV 2019	VGG19	62.8	101.8	7.7	12.7	88.7	154.8	75.2	299.5
NoisyCC [17]	NeurIPS 2020	Noisy-based	61.9	99.6	7.4	11.3	85.8	150.6	67.7	258.5
L2SM [15]	WACV 2021	HRNet-W48	62.4	98.4	7.2	11.1	104.7	173.6	-	-
KDMG [16]	T-PAMI 2021	Kernel-based	63.8	99.2	7.8	12.7	99.5	173.0	69.7	268.3
SASNet [18]	AAAI 2021	U-Net+VGG16	55.6	88.4	<u>6.3</u>	10.9	85.2	147.3	-	-
P2PNet [4]	AAAI 2021	VGG16	52.7	85.1	<u>6.3</u>	11.0	85.3	154.5	-	-
AutoScale [19]	IJCV 2022	VGG16	65.8	112.1	8.6	13.9	104.4	174.2	85.6	356.1
CLTR [13]	Sci. China 2022	Transformer	56.9	95.2	6.5	10.6	85.8	<u>141.3</u>	59.5	240.6
FIDTM [1]	TMM 2022	HRNet-W48	57.0	103.4	6.9	11.8	89.0	153.5	66.6	253.6
C-Trans [14]	ICCV 2023	Swin-T	66.1	105.1	9.3	16.1	97.2	168.5	74.9	295.6
AR-SAM [20]	ECCV 2024	SAM+VGG16	102.6	176.3	35.6	51.7	182.3	289.9	102.7	360.7
VMambaCC [21]	arXiv 2024	Mamba	<u>51.8</u>	<u>81.3</u>	7.4	12.4	88.4	144.7	<u>54.4</u>	<u>201.9</u>
CLIP-EBC [22]	arXiv 2024	ViT-B/16	56.6	94.6	8.1	15.1	87.7	159.3	50.2	126.5
P2R [23]	CVPR 2025	VGG16	51.0	79.6	6.1	9.8	83.2	138.1	58.8	253.1
AFT-Net	(Ours) 2025	HRNet-W48	53.6	101.3	6.1	<u>10.3</u>	<u>85.1</u>	<u>141.2</u>	66.5	250.8

The localization performance comparison is presented in Table 4. AFT-Net demonstrates highly competitive performance, particularly on the more challenging ShanghaiTech Part A dataset. At the strict threshold of $\sigma = 4$, it achieves the best Recall (59.2%) and a very competitive F1-score of 59.0%. More impressively, at the more lenient threshold of $\sigma = 8$, our method surpasses all listed competitors, securing the top position in F1-score (79.1%), Precision (79.2%), and Recall (78.9%). On the ShanghaiTech Part B dataset, AFT-Net continues to deliver robust results, achieving the second-best performance across all metrics at $\sigma = 8$ (F1: 81.8%) and the second-best Precision (63.6%) at $\sigma = 4$. Notably, when compared with other map-based methods, AFT-Net significantly outperforms LoBB and is highly competitive with FIDTM, demonstrating its superior ability to precisely locate individuals in diverse scenes.

Table 4. Comparison of localization performance with state-of-the-art methods on ShanghaiTech Part A and Part B. Metrics are reported under two different matching thresholds ($\sigma = 4$ and $\sigma = 8$). F, P, and R denote F1-score, Precision, and Recall, respectively. The best and second best results are in **bold** and underlined.

Method	Venue/Year	Manner	ShanghaiTech A						ShanghaiTech B					
			$\sigma = 4$			$\sigma = 8$			$\sigma = 4$			$\sigma = 8$		
			F(%)	P(%)	R(%)	F(%)	P(%)	R(%)	F(%)	P(%)	R(%)	F(%)	P(%)	R(%)
LCFCN [24]	ECCV 2018	Detection	32.2	42.3	26.0	54.6	74.3	43.1	-	-	-	-	-	-
LoBB [25]	CVPR 2019	Map-based	26.8	35.8	21.5	54.5	68.1	45.2	-	-	-	-	-	-
LSC [26]	T-PAMI 2020	Detection	32.6	32.3	32.9	62.9	62.0	63.6	29.7	29.8	29.9	56.7	55.9	57.5
P2PNet [4]	AAAI 2021	Point-based	40.6	41.5	39.8	74.6	76.2	73.1	-	-	-	-	-	-
CLTR [13]	ECCV 2022	Point-based	43.2	43.5	42.7	74.2	74.9	73.5	<u>63.3</u>	<u>62.7</u>	<u>63.9</u>	79.6	79.9	78.8
FIDTM [1]	TMM 2022	Map-based	<u>58.6</u>	59.1	<u>58.2</u>	77.6	78.2	77.0	64.7	64.9	64.5	83.5	83.9	83.2
APGCC [27]	ECCV 2024	Point-based	48.7	49.2	48.3	<u>78.4</u>	<u>79.1</u>	<u>77.7</u>	-	-	-	-	-	-
AFT-Net	(Ours) 2025	Map-based	59.0	<u>58.9</u>	59.2	79.1	79.2	78.9	63.4	62.5	63.6	<u>81.8</u>	<u>82.1</u>	<u>81.4</u>



Fig. 5. Qualitative comparison across various scenes from our benchmark. Each row represents a different scenario (Top: `indoor_base`, Middle: `indoor_light`, Bottom: `indoor_shadow`). For each scene, we show (a) the input image with ground-truth key-points, (b) the localization map from the baseline FIDTM, and (c) the map from our AFT-Net. Our method consistently produces superior results, especially under challenging lighting conditions.

4.5 Qualitative Analysis

To provide a more intuitive understanding of our method’s advantages, we present a wide range of qualitative results. Figure 5 showcases visual comparisons on samples from our `indoor_robustness` benchmark. The results clearly demonstrate our model’s superiority. In all cases, AFT-Net produces cleaner and more accurate localization maps with fewer false positives in glare and fewer

missed detections in shadows compared to the baseline, confirming its strong performance under diverse conditions.

5 Conclusion

This paper focused on solving the robustness problem of crowd localization under complex indoor lighting. We proposed AFT-Net, whose core is a novel Adaptive Focal Transform (AFT) mechanism. We also constructed the first benchmark dataset for this problem. Extensive experiments show that AFT-Net has significant robustness advantages in both counting and localization, and sets new state-of-the-art records.

Limitations and Future Work. While AFT-Net demonstrates strong robustness to common lighting variations, its performance on unseen, extreme noise types (e.g., severe motion blur or non-standard artistic lighting) has not been evaluated. Future work could explore extending the adaptive mechanism to handle a broader range of environmental perturbations. Furthermore, applying the AFT concept to video-based crowd tracking, which requires temporal consistency, presents a promising research direction.

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